Keya Prajapati

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.35
Vithal Vidyalaya	Class XII (GSEB)	2024	88.46%
Tejas Vidyalaya	Class X (GSEB)	2022	91.83%

SKILLS

Programming Languages: Python, C/C++, SQL

Core CS Fundamentals: Data Structures & Algorithms (Arrays, Linked Lists, Graph Traversal, Heaps), Object-Oriented Programming in C++ (Encapsulation, Inheritance, Polymorphism), SQL (Joins, Subqueries, Aggregate Functions)

Machine Learning: Supervised & Unsupervised Learning, Classification, Cross-Validation, Hyperparameter Tuning

Tools: Git, GitHub, MySQL, Matplotlib, NumPy, Pandas

PROJECTS

Global Cost-of-Living Predictor | Python, SQL | GitHub

- **Designed and populated a normalized SQLite database** collecting population, unemployment, and income data for **18 cities across 9 countries** (BLS, MoSPI, ONS, Destatis, INSEE, ABS, Numbeo), using a fact-table pattern to handle sparse, heterogeneous indicator availability.
- Engineered **PPP-adjusted income** and **log-scaled population** as derived features, correcting for currency and purchasing-power disparities and revealing a **4x gap** between nominal and PPP-adjusted income for Indian cities.
- Trained **Linear Regression and Random Forest** models using **Leave-One-Out cross-validation** (n=18), achieving **R^2=0.73**, with income contributing **89.5%** of feature importance in predicting the Cost-of-Living Index.
- Validated pipeline reliability with a **25-test pytest suite** covering database integrity and currency-conversion correctness, and quantified per-city data quality via a custom **data confidence score**.

Disk-Backed Virtual Memory & RAID Storage in xv6 | GitHub

- Redesigned xv6's **in-memory swap** into a **disk-backed virtual memory system**, introducing a **priority-aware request queue** and a **seek-latency model** to simulate realistic disk access costs
- Implemented **FCFS and SSTF disk scheduling algorithms** with **scheduler-aware tie-breaking**, reducing average seek time under high I/O contention
- Built a full **software RAID layer (RAID 0/1/5)** from scratch, supporting **degraded-mode reconstruction** and tightly integrated with **virtual memory, page fault handling, and fork()** for transparent process-level access
- Validated correctness and performance through **custom stress-test experiments**, tracking **swap activity and disk I/O statistics** across varying workloads

Macro Dashboard & Recession Signal Tracker | python | github

- Built an **end-to-end pipeline** pulling GDP, CPI, unemployment, and Treasury yield data (**1976-2026**, 603 monthly observations) from the FRED API, cleaning and aligning series of differing frequencies into a **unified monthly dataset**.
- Engineered the **10Y-2Y yield curve spread** and **year-over-year inflation rate** as derived signals, validating the **yield curve inversion** as a recession predictor against NBER recession dates (**5 of 6 recessions since 1976** preceded by an inversion).
- Designed **visualizations with recession-period overlays** to reveal relationships between unemployment, inflation, and yield curve behavior across **five decades**.
- Interpreted findings against macroeconomic theory (**Phillips curve**, **stagflation**), identifying where historical data **confirmed or broke** from classical models.

Loan Approval Prediction | Python | Github

- Engineered an end-to-end **binary classification** pipeline comparing **Logistic Regression, KNN, Decision Tree, Random Forest, and SVM**, selected via **5-fold stratified cross-validation**
- Built a **leakage-free preprocessing pipeline** using ColumnTransformer for **imputation, feature scaling, and one-hot encoding**, and tuned the top model with **GridSearchCV**
- Achieved **F1 score of 0.90** and **ROC-AUC of 0.84** on held-out test data, prioritizing **precision and recall over accuracy** to account for class imbalance (69:31 split)
- Supplemented supervised analysis with **PCA** for dimensionality reduction and **K-Means clustering** for exploratory customer segmentation

RELEVANT COURSES

Computer Science:

Introduction to Programming, Data Structures & Applications, Operating Systems I & II

Mathematics & Statistics

Calculus I & II, Elementary Linear Algebra, Differential Equations, Transform Techniques, Complex Variables, Introduction to Probability, Introduction to Statistics

POSITIONS OF RESPONSIBILITY

Design Core: Ek Bharat Shreshtha Bharat Club

EXTRACURRICULAR

3D Art Portfolio

3D Art & Visualization -- Personal pursuit of cinematic and automotive 3D rendering using **Blender** and **Unreal Engine 5**, with work published on **ArtStation**